Fiscal Year:	FY 2008	**Task Last Updated:**	FY 03/30/2009
PI Name:	Oman, Charles M. Ph.D.		
Project Title:	Human Orientation and Sensory-Motor Coordination in Prolonged Weightlessness (ISS HRF Expt. E085)		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02139-4301	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	96-OLMSA-01
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:	ISS		
Flight Assignment:	This funding covers his support as a Co-I to the ESA managed flight study.		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Liu, Andrew (MIT)		
Grant/Contract No.:	NNJ05HD39G		
Performance Goal No.:			
Performance Goal Text:			
	This International Space Station Utilization Flight Experiment (E085), is being conducted as part of Human Research Facility (HRF) Experiment: Visuomotor and Orientation Investigations in Long Duration Astronauts (VOILA). VOILA merges and extends into long duration flight two investigations originally developed for the 1998 Neurolab space shuttle mission. The science team includes investigators from the United States, Canada, France, and Italy. This experiment addresses the primary area of Space Physiology and Countermeasures. The Neurovestibular Adaptation sections of the NASA Critical Path Roadmap identift several risks of long-duration spaceflight requiring scientific investigation and countermeasure development. The information from our experiments addresses several critical questions identified within the highest ranked risk (#33), Disorientation and Inability to Perform Landing, Egress, or Other Physical Tasks, Especially During/After G-Level Changes. Our experiments will provide quantitative information useful for understanding the underlying mechanisms of the effects of weightlessness on disorientation, visual reorientation		

Task Description:	illusions (the downs), inversion illusions, EVA acrophobia, motion sickness, and human sensorimotor coordination (Critical Question 9.09) involved with reaching and catching, and the development of practical countermeasures (Critical Question 9.08) such as preflight visual orientation training. No scientific data on these phenomena is yet available from long duration space flights, and our experiment provides the opportunity to rapidly move from CRL0 to CRL5 in these areas. In terms of likelihood and consequences, spatial disorientation is considered the highest(red/one) neurovestibular risk. Our experiments also provide information relevant to the secondary area of human behavior and performance within ineffective habitat critical path risk (#20), as relates to spacecraft interior layout and work area design. On earth, gravity provides a reference that influences how we recognize objects in our field of view and perceive their shape and orientation. It also influences our expectations of how objects will behave when thrown or dropped. The orientation of familiar visual objects and surfaces in turn influence our own perceived orientation. Gravity also helps the central nervous system align the various frames of reference used in movement control. All of these processes are fundamentally altered in weightlessness, as evidenced by the visual reorientation, inversion, and proprioceptive illusions frequently reported in-orbit by astronauts, and also by the surprisingly long lasting after effects seen, particularly after prolonged space flight. The experiment utilizes the HRF Rack 2 graphics workstation as a tool to study these processes during and after orbital flight. In VOILA, restrained and free-floating subjects wear a wide field of view, color stereo head mounted display, and view controlled visual scenes in five different test paradigms which assess: (1) the influence of scene symmetry, rotation, haptic cues, and expected orientation on static and dynamic self tilt (Virtual Tilting and Tumbling Room Tests)and visual reorientation illusions (VRIs); (2) the onset of x-axis illusory linear self-motion without haptic cues (Linear Vection Test); (3) the effect of perceived orientation on visual object recognition and shape recognition (Object Recognition Tests); (4) whether information used in grasping remembered objects is stored in head fixed, body fixed, or exocentric reference frames (Virtual Grasping Test); and (5) how the timing of catching movements depends on anticipation of downward acceleration (Virtual Catching Test). The 15-45 minute long test modules are based on existing 1-G paradigms, and can be selected and performed by an astronaut in an automated fashion using Session Manager software. Two preflight, three inflight, and three postflight performances of each test are planned on each ISS increment. An engineering team from the MIT Center for Space Research is designing the suite of virtual reality hardware and software for the workstation, including a head mounted display, hybrid inertial-optical 6 degree-of-freedom head tracker, a subject input device, and a subject restraint system. The latter consists of a folding chair-like body restraint, and a pole equipped with a swivel joint for free floating experiments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Insight into how astronauts adapt to microgravity should provide valuable information into the normal functioning of the human nervous system. Strategies used by human subjects to orient and navigate in three dimensions and compensate for the lack of gravity may also suggest how the neural disorders mentioned above might be treated. There is also currently considerable interest in the development of new methods for evaluating a patient's ability to use visual and proprioceptive cues to maintain balance and orientation, and for improving motor control function via rehabilitative training. Portable head mounted displays and head and hand tracking systems, akin to those used in this experiment, may prove useful for diagnostic testing and rehabilitative training.
Task Progress:	This report summarizes the activities of the MIT science team from 5/1/2005 to 4/30/2008 supported by NASA Grant NNJ05HD39G. In 1996, NASA selected an MITCNES proposal “Human Orientation and Sensory-Motor Coordination in Prolonged Weightlessness” an ISS basic neuroscience experiment submitted in response to NRA96- OLMSA-01A The international experiment was renamed VOILA (Visuomotor and Orientation Investigations in Long-duration Astronauts) and assigned NASA flight operations number E085 and ESA number E502. Flight hardware – a virtual reality hardware suite for the HRF graphics workstation - was developed by the MIT Center for Space Research via contract NAS9-00097, working in collaboration with ESA/CNES. The international development effort proceeded through the Critical Design Review stage. However, as a result of the Columbia accident, and a restructuring of the timeline and priorities for NASA science utilization of ISS, the engineering development of the experiment was discontinued. Extensive preflight science and engineering was accomplished. Most all prototype flight hardware was transferred to ESA/CNES. The present grant provided 3 years of science support for the MIT VOILA science team to continue development of the VOILA science protocols as Co-Investigators on the ESA-led E507 experiment, which remained in development through July of 2007. Scientific progress: Between 2002 and 2004, we conducted a series of experiments on three parabolic flight campaigns sponsored by the CNES to investigate changes in the perception of illusory self-motion (vection) during microgravity exposure and on the ground in 1g. These ground-based experiments were carried out to also support the development of the in-flight VECTION protocols for VOILA. The experiments were in part an outgrowth of the 2004 Master’s thesis work on linear vection of Mr. Kevin Duda. After the July 2007 teleconference with Dr. Joe McIntyre, in which it was reported that the VOILA experiment had moved significantly down the ESA queue, the science team refocused its efforts on finishing the analysis of the parabolic flight data and preparing a manuscript for publication. During this time, the interim analysis has been reported in the poster session at the 2008 NASA Human Research Program Investigator Workshop (HRPIW). As of June 2008, a draft manuscript has been completed , and will be submitted to the Journal of Vestibular Research for publication.
Bibliography Type:	Description: (Last Updated: 12/18/2024)
Abstracts for Journals and Proceedings	Liu AM, Oman CM, Duda K, Natapoff A, McIntyre JA. "Human Orientation and Sensory-Motor Coordination in Prolonged Weightlessness." NASA Human Research Program Investigators’ Workshop, Houston, TX, Feb 4, 2008. NASA Human Research Program Investigators’ Workshop, Houston, TX, Feb 4, 2008. , Feb-2008

Articles in Peer-reviewed Journals	Liu AM, Duda KR, Oman CM, Natapoff A. "The perception of visually induced linear self-motion during parabolic flight." To be submitted to Journal of Vestibular Research, November 2008. , Nov-2008